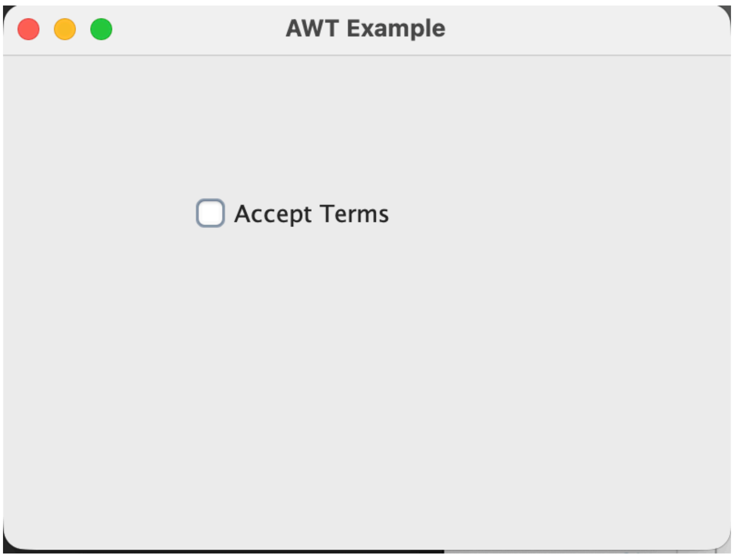
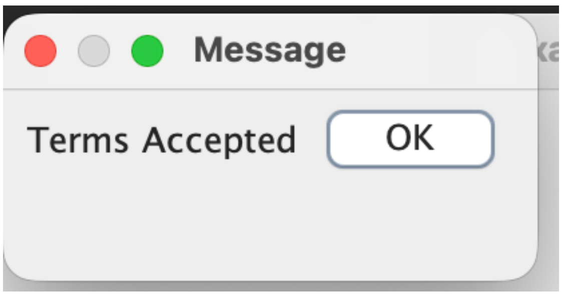
Q1. Design a GUI application first using AWT and then implement the same application using Swing components. The application should contain a combo box or checkbox, and upon selecting an option, a dialog box should be displayed asking the user whether they accept the terms and conditions.

Lab_11 > awt.java > awt > main(String[])

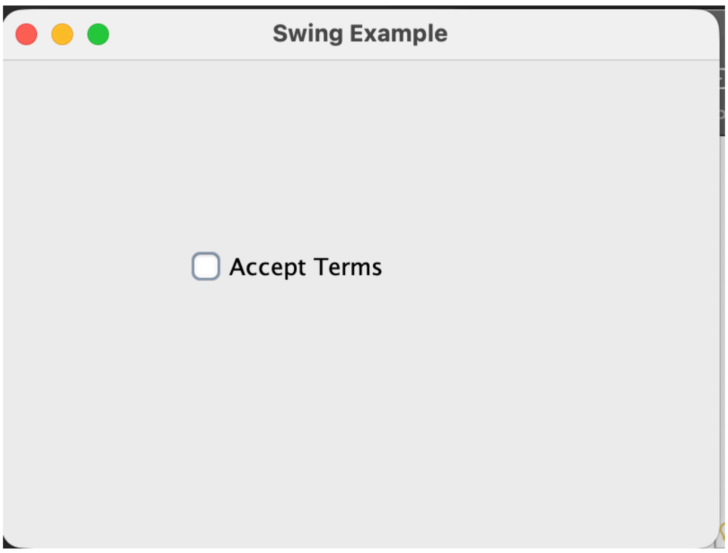
```
/*abstract window toolkit It is Java's old GUI library used to create:
* Window
* Button
* Checkbox
* TextField
* Dialog Box
AWT components depend on operating system.*/
package Lab_11;
import java.awt.*;
import java.awt.event.*;
public class awt {
    Run | Debug
    public static void main(String[] args) {

        Frame f = new Frame(title: "AWT Example");//window banata h ye

        Checkbox cb = new Checkbox(label: "Accept Terms");
        cb.setBounds(x: 100, y: 100, width: 150, height: 30);
        cb.addItemListener(new ItemListener() {
            public void itemStateChanged(ItemEvent e) {
                if (cb.getState()) {
                    Dialog d = new Dialog(f, title: "Message", modal: true);
                    d.setLayout(new FlowLayout());  //Components arranged left to right.
                    d.add(new Label(text: "Terms Accepted"));
                    Button b = new Button(label: "OK");
                    b.addActionListener(a -> d.dispose());   //dispose()close window.
                    d.add(b);
                    d.setSize(width: 200, height: 100);
                    d.setVisible(b: true);  //Makes dialog visible.
                }
            }
        });
        f.add(cb);
        f.setSize(width: 400, height: 300);
        f.setLayout(mgr: null);
        f.setVisible(b: true);
        f.addWindowListener(new WindowAdapter() { //Handles close button event.
            public void windowClosing(WindowEvent e) {
                f.dispose();
            }
        });
    }

```

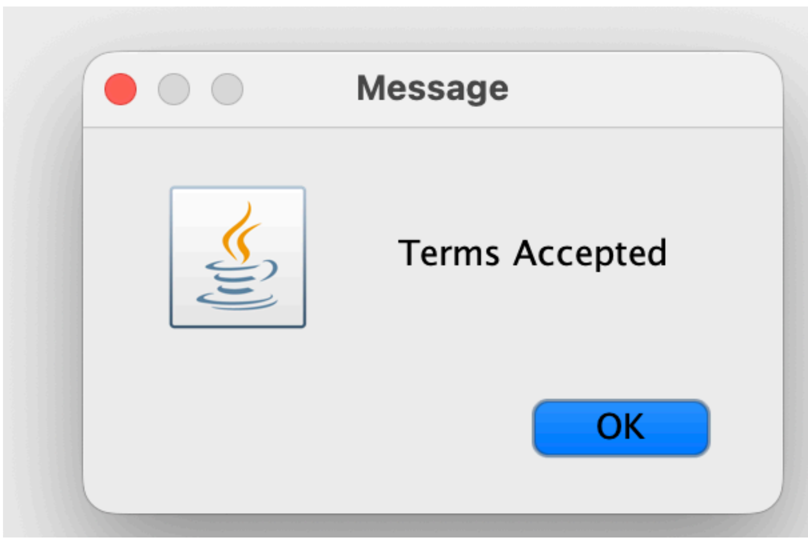




Lab_11 > Swing.java > Swing

```java
//Advanced version of awt
package Lab_11;
import javax.swing.*;
import java.awt.event.*;
public class Swing {
    Run | Debug
    public static void main(String[] args) {
        JFrame f = new JFrame(title: "Swing Example");
        JCheckBox cb = new JCheckBox(text: "Accept Terms");
        cb.setBounds(x: 100, y: 100, width: 150, height: 30);
        cb.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                if (cb.isSelected()) {
                    JOptionPane.showMessageDialog(f,
                            message: "Terms Accepted");
                }
            }
        });
        f.add(cb);
        f.setSize(width: 400, height: 300);
        f.setLayout(manager: null);
        f.setVisible(b: true);
        f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
}
```





Q2. Develop a GUI-based Login and Registration System using Swing, Event Handling, and JDBC. The GUI should contain appropriate components for user registration, including fields for Name, Username, Password, Gender, Hobbies, etc., along with Register, Login, and Clear buttons. On clicking the Register button, the entered details should be stored in the database, and a dialog box displaying “User Registered” should appear. On clicking the Login button, the system should validate the entered username and password from the database. If the credentials are valid, a “Login Successful” dialog box should be displayed; otherwise, an “Invalid User” dialog box should appear. The Clear button should reset all input fields.

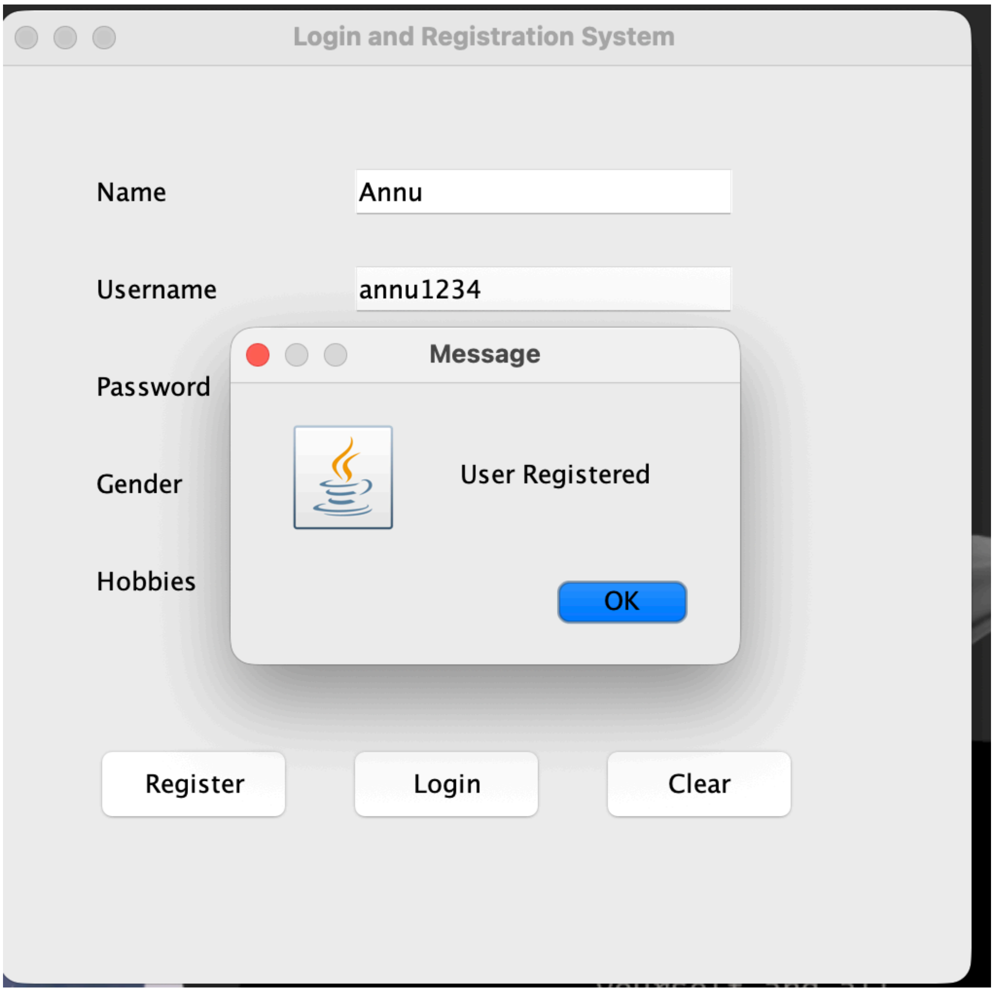
Lab_11 > LoginSystem.java > LoginSystem > LoginSystem()

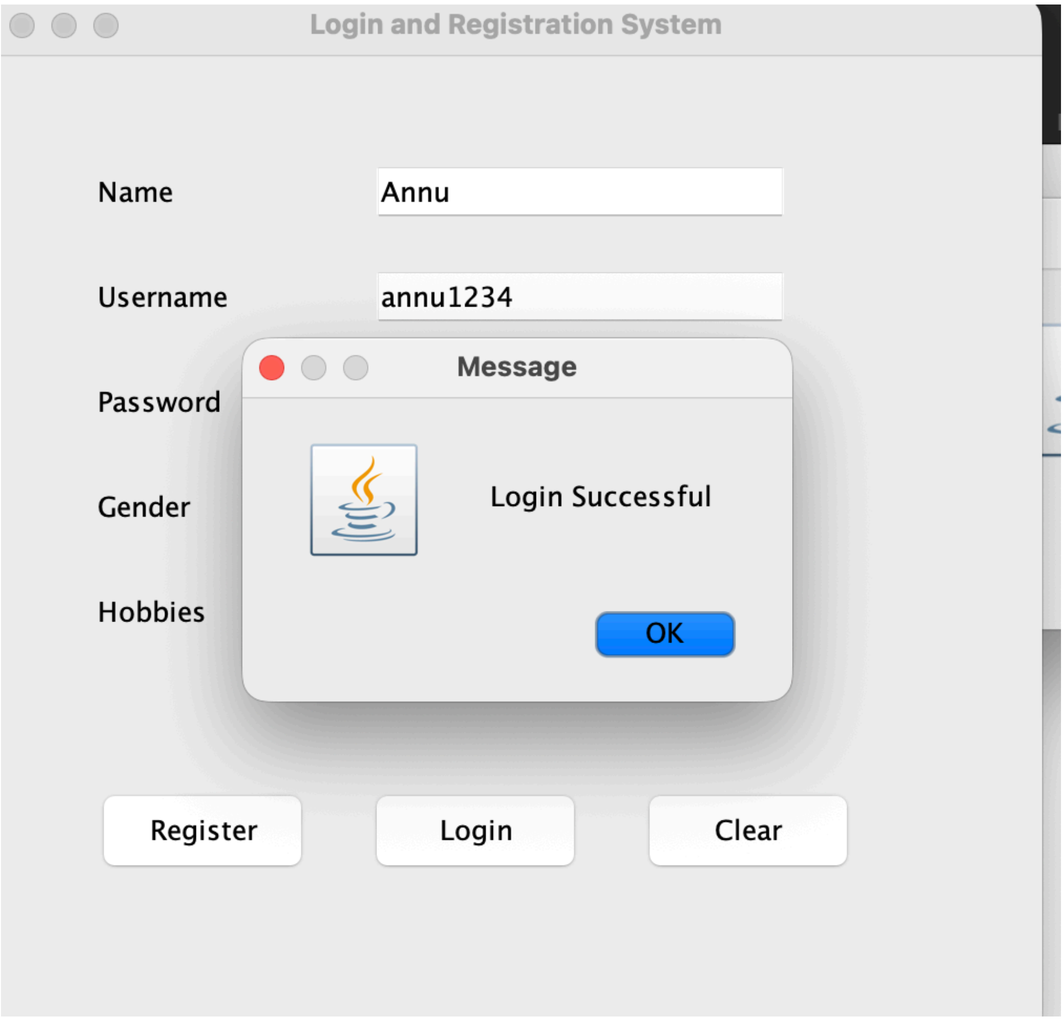
```java
package Lab_11;
import javax.swing.*;
import java.awt.event.*;
import java.sql.*;
public class LoginSystem extends JFrame implements ActionListener {
    JTextField t1, t2;
    JPasswordField t3;
    JRadioButton male, female;
    JCheckBox music, sports;
    JButton register, login, clear;
    Connection con;
    LoginSystem() {
        // Window
        setTitle(title: "Login and Registration System");
        setSize(width: 500, height: 500);
        setLayout(manager: null);
        // Name
        JLabel l1 = new JLabel(text: "Name");
        l1.setBounds(x: 50, y: 50, width: 100, height: 30);
        add(l1);
        t1 = new JTextField();
        t1.setBounds(x: 180, y: 50, width: 200, height: 30);
        add(t1);
        // Username
        JLabel l2 = new JLabel(text: "Username");
        l2.setBounds(x: 50, y: 100, width: 100, height: 30);
        add(l2);
        t2 = new JTextField();
        t2.setBounds(x: 180, y: 100, width: 200, height: 30);
        add(t2);
        // Password
        JLabel l3 = new JLabel(text: "Password");
        l3.setBounds(x: 50, y: 150, width: 100, height: 30);
        add(l3);
        t3 = new JPasswordField();
        t3.setBounds(x: 180, y: 150, width: 200, height: 30);
        add(t3);
        // Gender
        JLabel l4 = new JLabel(text: "Gender");
        l4.setBounds(x: 50, y: 200, width: 100, height: 30);
        add(l4);
        male = new JRadioButton(text: "Male");
        male.setBounds(x: 180, y: 200, width: 80, height: 30);
        female = new JRadioButton(text: "Female");
        female.setBounds(x: 280, y: 200, width: 100, height: 30);
        ButtonGroup bg = new ButtonGroup();
        bg.add(male);
        bg.add(female);
        add(male);
        add(female);
        // Hobbies
        JLabel l5 = new JLabel(text: "Hobbies");
        l5.setBounds(x: 50, y: 250, width: 100, height: 30);
        add(l5);
        music = new JCheckBox(text: "Music");
        music.setBounds(x: 180, y: 250, width: 100, height: 30);
        sports = new JCheckBox(text: "Sports");
        sports.setBounds(x: 280, y: 250, width: 100, height: 30);
        add(music);
        add(sports);
        // Buttons
        register = new JButton(text: "Register");
        register.setBounds(x: 50, y: 350, width: 100, height: 40);
        login = new JButton(text: "Login");
        login.setBounds(x: 180, y: 350, width: 100, height: 40);
        clear = new JButton(text: "Clear");
```

```
    LoginSystem() {
        login = new JButton(text: "Login");
        login.setBounds(x: 180, y: 350, width: 100, height: 40);
        clear = new JButton(text: "Clear");
        clear.setBounds(x: 310, y: 350, width: 100, height: 40);
        add(register);
        add(login);
        add(clear);
        // Events
        register.addActionListener(this);
        login.addActionListener(this);
        clear.addActionListener(this);
        // Database Connection
        try {
            Class.forName(className: "com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                    url: "jdbc:mysql://localhost:3306/testdb",
                    user: "annu",
                    password: ""
            );
            System.out.println(x: "Database Connected");
        } catch (Exception e) {
            System.out.println(e);
        }
        setVisible(b: true);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }
    public void actionPerformed(ActionEvent e) {
        String name = t1.getText();
        String username = t2.getText();
        String password = new String(t3.getPassword());
        String gender = "";
        if (male.isSelected()) {
            gender = "Male";
        } else if (female.isSelected()) {
            gender = "Female";
        }
        String hobbies = "";
        if (music.isSelected()) {
            hobbies += "Music ";
        }
        if (sports.isSelected()) {
            hobbies += "Sports";
        }
        // REGISTER
        if (e.getSource() == register) {
            try {
                PreparedStatement ps = con.prepareStatement(
                        sql: "INSERT INTO users VALUES (?, ?, ?, ?, ?)"
                );
                ps.setString(parameterIndex: 1, name);
                ps.setString(parameterIndex: 2, username);
                ps.setString(parameterIndex: 3, password);
                ps.setString(parameterIndex: 4, gender);
                ps.setString(parameterIndex: 5, hobbies);
                int x = ps.executeUpdate();
                if (x > 0) {
                    JOptionPane.showMessageDialog(
                            this,
                            message: "User Registered"
                    );
                }
            } catch (Exception ex) {

                System.out.println(ex);
            }
```

```java
                System.out.println(ex);
            }
        }
        // LOGIN
        if (e.getSource() == login) {
            try {
                PreparedStatement ps = con.prepareStatement(
                        sql: "SELECT * FROM users WHERE username=? AND password=?"
                );
                ps.setString(parameterIndex: 1, username);
                ps.setString(parameterIndex: 2, password);
                ResultSet rs = ps.executeQuery();
                if (rs.next()) {
                    JOptionPane.showMessageDialog(
                            this,
                            message: "Login Successful"
                    );

                } else {
                    JOptionPane.showMessageDialog(
                            this,
                            message: "Invalid User"
                    );
                }
            } catch (Exception ex) {
                System.out.println(ex);
            }
        }
        // CLEAR
        if (e.getSource() == clear) {
            t1.setText(t: "");
            t2.setText(t: "");
            t3.setText(t: "");
            male.setSelected(b: false);
            female.setSelected(b: false);
            music.setSelected(b: false);
            sports.setSelected(b: false);
        }
    }
    Run | Debug
    public static void main(String[] args) {
        new LoginSystem();
    }
}
```

Lab:-11


Login and Registration System
Name
Annu
Username
annu1234
Password
Gender
Hobbies
Message
User Registered
OK
Register
Login
Clear


Login and Registration System
Name
Annu
Username
annu1234
Password
Gender
Hobbies
Message
Login Successful
OK
Register
Login
Clear